

Salah Uddin Selim

CSE Student • Backend Developer • AI/ML • Data Analyst • SQA

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EDUCATION

B.Sc. in Computer Science & Engineering

United International University

2022 – Present

Dhaka, Bangladesh

CGPA: **3.74** / **4.00** • Credits completed: **126** / **137** • Relevant coursework: DSA, OOP, AI, ML, DB, Networks, SE, SQA, DM

HSC — Pirganj Govt. College, Thakurgaon (2018–2020) **GPA 5.00/5.00** • **SSC** — Pirganj Pilot High School, Thakurgaon (2013–2018) **GPA 5.00/5.00**

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, PHP, SQL

Embedded / IoT: Arduino, C++ (Arduino IDE), Sensor Integration

Frontend: HTML, CSS, Tailwind CSS, Bootstrap, JavaFX

Tools: Git, GitHub, VS Code, IntelliJ IDEA, Vercel, Postman, Xampp

Backend: PHP, REST APIs, MySQL

Concepts: OOP, Multithreading, Socket Programming, Survival Analysis

Research: Medical AI, Bioinformatics, Multi-Omics, Computer Vision, XAI

PROJECTS

LeukemiaNet-XAI — Explainable Leukemia Classification (IEEE/Scopus target)

2024 – Present

- Built a multi-backbone deep-learning pipeline (EfficientNetV2-L, ConvNeXtV2, Swin-Base + cross-attention ensemble) classifying four leukemia subtypes (ALL, AML, CML, CLL) from blood-smear images.
- Deployed GradCAM++, SHAP, and Integrated Gradients for clinical interpretability; used Optuna for automated hyperparameter optimisation.
- Manuscript in preparation for an IEEE/Scopus-indexed venue; presented at FYDP-II defence. *Stack: Python, TensorFlow/Keras, scikit-learn, Optuna*

Multi-Omics Cancer Survival Prediction System

2024 – Present

- Integrated TCGA Pan-Cancer Atlas gene expression, CNV, and methylation data; reduced ~20,000 features to a clinically interpretable subset via feature-selection modules.
- Cox-PH and XGBoost-Survival models exceeded published C-index baselines on held-out cohorts. *Stack: Python, lifelines, scikit-survival, XGBoost, Pandas*

Multilevel Puzzle Solving Game — 6th Place, UIU Inter-Team Competition

Spring 2025

- Architected a networked team-based desktop app (5 puzzle mini-games) with Java Socket real-time group chat, multithreading, shared level-unlock mechanics, and MySQL persistence.
- *Stack: Java, JavaFX, Scene Builder, Java Sockets, MySQL* 🌐 [PuzzleSolving-Game](#)

TournyMate — Tournament Management System

Fall 2024

- Delivered a full-stack web platform with automated match scheduling, real-time live scores, role-based access control (admin/player), and responsive UI across devices.
- *Stack: PHP, MySQL, HTML, CSS, Bootstrap, JS* 🌐 [tourny_mate](#)

Automated Fish Pond — IoT Water Quality Monitor

Summer 2024

- Integrated five sensor types (temperature, pH, turbidity, gas, ultrasonic) with Arduino; programmed C++ decision-tree logic to autonomously trigger water exchange, pH correction, and feed dispensing.
- *Stack: C++, Arduino Uno, servo motors, L298N* 🌐 [Automated-Fish-Pond](#)

AI Voice Assistant & Player Information System

2023

- **Voice Assistant:** Python app using SpeechRecognition + Google Search API for natural-language OS automation and web queries with < 2 s latency.
- **Player Info System:** Menu-driven C app with file I/O, struct arrays, and statistical aggregations for Bangladesh Cricket Team data.
- 🌐 [Voice-Assistance](#) 🌐 [Player-Information-System](#)

ACHIEVEMENTS

- 6th Place — UIU Software Competition (Spring 2025)
- CGPA **3.74** / **4.00** across 126 credits at UIU
- GPA **5.0** / **5.0** — Both HSC and SSC national exams

RESEARCH INTERESTS

Medical AI • Explainable AI (XAI) • Bioinformatics
Multi-Omics Integration • Survival Analysis
Computer Vision • IoT Systems